MIL-S-55616A(CR)
11 August 1980
Superseding
MIL-S-55616(EL)
14 June 1968

MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT
S-392( )/TTC-29

This specification is approved for use by the Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of the Central Office Telephone Manual AN/TTC-29( ) which is capable of switching facilities for 60 circuits.

1.2 Description.- Shelter, Electrical Equipment S-392( )/TTC-29 hereinafter called the shelter facility, includes a modified S-250( )/G shelter and all racks, brackets, shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TTC-29( ).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, & Packaging, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research and Development Command, ATTN: DRDCO-PE-ES-1, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5805

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MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys and Titanium, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable, for Structures, Excluding Armor
MIL-S-55507	Shelter, Electrical Equipment (Empty) Packaging
MIL-C-55427	Cable Assembly, Radio Frequency General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-461	Characteristics Requirements for Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

ELECTRONIC COMMAND

DL-SC-A-595488	Shelter Electrical Equipment S-392( )/TTC-29

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol shall be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- The shelter facility shall be constructed in accordance with the drawings as listed on DL-SC-A-595488.

3.3 Electrical installation.-

3.3.1 Power system and components installation.- (See 4.6.1) Installation and wiring of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system and wiring components constituting a three wire electrical power distribution system) shall be in accordance with the drawings unless otherwise stated. The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- There shall be point-to-point circuit continuity on all power-line circuits and the resistance of the continuous circuit shall not exceed 3 ohms when tested in accordance with 4.6.1 and 4.6.1.1.

3.3.1.2 Dielectric breakdown voltage.- The dielectric strength of insulation of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.6.1 and 4.6.1.2.

3.3.1.3 Insulation resistance.- The resistance of insulation between any two of three circuit conductors and also between each of the two ungrounded power-line conductors and shelter facility ground terminal shall not be less than 100 megohms when tested in accordance with 4.6.1 and 4.6.1.3.

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3.3.2 Power distribution and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by the power system schematic plate, when tested in accordance with 4.6.2.

3.3.3 Signal (audio) system wiring and components installation.- (See 4.6.3) Installation and wiring of signal wires, cables and wiring components (including binding posts, connectors, plugs, jacks, receptacles, switches, terminal strips, terminal blocks, and wiring devices) shall be in accordance with the drawings unless otherwise stated. The completed in-place signal system installation shall meet the following requirements:

3.3.3.1 Continuity of conductors.- There shall be circuit continuity between both end terminations of all signal - line conductors and the resistance of the continuous circuit shall not exceed 5 ohms when tested in accordance with 4.6.3.1.

3.3.3.2 Dielectric breakdown voltage.- The dielectric strength of insulation of all signal wires, cables and components shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds when tested in accordance with 4.6.3.2.

3.3.3.3 Insulation resistance.- The resistance of insulation between each signal-line conductor and all other signal-line conductors connected to ground shall not be less than 100 megohms when tested in accordance with 4.6.3.3.

3.3.4 Cable assemblies.- (See 4.4)

3.3.4.1 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull-out requirements of MIL-C-3885.

3.3.5 Suppression of electromagnetic interference.- The motors in the ventilating equipment used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4)

3.4 System operation.- With all the operational equipment of the AN/TTC-29( ) assemblage installed, the operational requirements of all power and audio cable assemblies, switches, and patch panel, shall be met when tested in accordance with 4.7.

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3.5 Air tightness.- The shelter facility shall show no evidence of air leakage through sealed seams, joints or rivits of exterior surface modified per DL-SC-A-595488 when tested per 4.8.

3.6 Railroad transportation.- The shelter facility when assembled as the AN/TTC-29 without any extra tieing, taping, etc., shall be capable of being loaded, blocked, and braced on a flat-car for shipment by rail and shall be subjected to and withstand the test of paragraph 4.9 without sustaining any buckling, cracking, delamination, or other permanent deformation.

3.7 Drops, flat and rotational.- The shelter facility when assembled as the AN/TTC-29 shall be capable of withstanding drops of 18 inches onto concrete. After all drop tests of 4.10.1 and 4.10.2 there shall be no buckling, cracking, delamination or other permanent deformation.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.11. During and after the test there shall be no leakage into the facility or its walls, floor, roof or doors.

3.9 Fording.- The shelter facility shall be capable of withstanding the tests of 4.12. During and after the test there shall be no leakage into the facility or its walls, floor, roof or doors.

3.10 Mounting inserts.- All mounting inserts shall be be installed without the use of any auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.13. There shall be no damage or permanent distortion to the shelter mounting surface or insert head or threads and the inserts shall not have rotated from its original position. Insert threads shall be free of paint.

3.11 Cleaning.- (See 3.22)

3.11.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After

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assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.11.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.13 Marking.- (See 4.4)

3.13.1 General.- Markings shall conform to Specification MIL-M-13231 and the drawings.

3.13.2 Nameplates.- Shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with the operational equipments or dummy loads in place. The location of the centers-of-gravity shall be determined by the contractor using the first article of production.

3.13.4 Interior and exterior marking.- All interior and exterior markings shall be as defined on the drawings.

3.14 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Seal, water vapor.- All joints of the framing for the exhaust blowers, power, signal, and video entrance boxes, and any other framed openings shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges

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shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one-part sealant, Silicon Construction Sealant SE-1204, as made by General Electric Corporation or Silastic, 732 as made by Dow Corning, or equal using the manufacture recommended primer.

3.16 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.-

3.17.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of eletrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operational shall be undamaged and parts fastened thereto shall not have become loosened.

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3.18 Tropicalization of material.- Canvas, webbing, wood, etc shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials. to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2e)

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is on the drawings. Inert-gas-shielded arch welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B and MIL-W-45205 CL.B as applicable. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

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3.20 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires and connections.

b. To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cable may be connected of disconnected.

c. To facilitate field repair of broken or cut wires.

d. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insultation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.4 and 4.14)

3.20.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.15). Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any specified dimension, rating, characteristic, etc.

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3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.2 Construction
3.3 Electrical installation
3.10 Mounting inserts
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal, water vapor
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and cabling
3.21 Interchangeability

4. QUALITY ASSURANCE PROVISIONING

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

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(2) Quality conformance inspection for packaging. (See 4.18)

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First article units.- The contractor shall furnish one (1) first article unit complete S-392( )/TTC-29.

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I - First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	-	4.4
2. Interchangeability	3.21	4.15
3. Mounting insert	3.10	4.13.1
4. Solderless-terminal lug connectors	3.20.4	4.14
5. Visual and mechanical	3.22	4.16
6. Power systems and components installation	3.3.1	4.6.1
7. Power distributional and operational function	3.3.2	4.6.2
8. Signal (audio) sytem wiring and components installation	3.3.3	4.6.3
9. System operation	3.4	4.7
10. Air-tightness test	3.5	4.8
11. Rail transportation	3.6	4.9

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12.	Flat drop	3.7	4.10.1
13.	Rotational drop	3.7	4.10.2
14.	Watertightness	3.8	4.11
15.	Fording	3.9	4.12
16.	Repeat 6. Power systems and components installation	3.3.1	4.6.1
17.	Repeat 7. Power distributional and operational function	3.3.2	4.6.2
18.	Repeat 8. Signal (audio) system wiring and components installation	3.3.3	4.6.3
19.	Repeat 10. System operation	3.4	4.7

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

Item	Where required
Cable assemblies	3.3.4
Suppression, EMI	3.3.5
Finish	3.12
Marking	3.13
Welding	3.19
Terminals	3.20.4

4.5 Quality conformance inspection of equipment before packaging.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

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4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in the order listed in Table II.

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % Def Major	Minor
Visual and mechanical	3.22	4.16	1.0	4.0
Electrical installation				**
Power systems and component installation	3.3.1	4.6.1		
Continuity of conductors	3.3.1.1	4.6.1.1		
Dielectric breakdown voltage	3.3.1.2	4.6.1.2		
Insulation resistance	3.3.1.3	4.6.1.3		
Power distributional and operational function	3.3.2	4.6.2		
Signal (audio) system wiring and components installation	3.3.3	4.6.3		
Continuity of conductors	3.3.3.1	4.6.3.1		
Dielectric breakdown voltage	3.3.3.2	4.6.3.2		
Insulation resistance	3.3.3.3	4.6.3.3	1.0 for the entire group of power and signal	

** All electrical defects are major.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting insert inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

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4.5.2.1 Group B sampling plans.- The group B sampling plans, for the AQL's listed in Table III shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order listed in Table III.

TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL
1. Mounting inserts	3.10	4.13.2	4.0%
2. Interchangeability	3.21	4.15	4.0%
3. Air tightness	3.5	4.8	2.5%

"NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in 4.5.2.2."

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Drops, flat and rotational	3.7	4.10
Solderless-terminal lug connection	3.20.4	4.14
System operation	3.4	4.7
Watertightness	3.8	4.11
Fording	3.9	4.12

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NOTE: Units subjected to Group C Drop Tests shall be re-subjected to the Air Tightness Test prior to the Watertightness and Fording Tests.

4.5.3.1 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimpling the terminal lugs in production.

4.5.3.2 Sampling for inspection of drop tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the railroad transportation test of 4.5.3.3. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.3 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the Government.

a. Mounting inserts
b. System operation
c. Railroad transportability

4.5.3.4 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Electrical installation tests.- The tests of 4.6.1 through 4.6.2.3 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to the shelter facility shall be 115 VAC, 60 cycles, single phase.

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4.6.1 Power system and components installation test.- Prior to conducting power wiring and components test of 4.6.1.1, 4.6.1.2 and 4.6.1.3, the following shall be accomplished:

a. Remove or disconnect the following electrical loads:

(1) Fluorescent lamps and lamp starters.

(2) Incandescent cold start lamps.

(3) Power distribution box neon indicator lamps.

(4) Power indicator neon lamp.

(5) Blower.

(6) Heater.

(7) Air conditioner.

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeters terminals. Position or insulate disconnected leads so that they do not short to each other or to ground.

4.6.1.1 Continuity of conductors.- Continuity measurements which satisfy the requirements 3.3.1.1 shall be made on each power-line circuit as indicated by the following paragraphs using an aural or visual indicating continuity test instrument and suitable test jig. The instrument shall produce a test current between 0.2 and 2.0 ampers AC when circuit continuity exists.

a. Place the following circuit controls in the ON position:

(1) Main circuit breaker.

(2) Branch circuit breakers.

(3) Fluorescent lamp switch.

(4) Incandescent cold start switch.

(5) Bypass/Blackout switch.

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(6) Night light switch.

(7) Blower switch.

b. Measure continuity from pin 1 (of power IN connector J1 in power entrance box) to the following: brass contact on each receptacle; black wire termination at each fluorescent lamp; and black wire termination at each incandescent lamp.

c. Place all controls listed under procedure 4.6.1.1a in OFF position, except main circuit breaker.

d. Measure continuity from pin 2 (of power IN connector in power entrance box) to the following: silver contacts on each receptacle; white wire termination at each fluorescent lamp and white wire termination at each incandescent lamp.

e. Place main circuit breaker in OFF position.

f. Measure continuity from pin 4 (ground) (of power IN connector) to each ground terminal on receptacles.

g. Measure continuity from pin 4 (ground) (of power IN connector) to shelter ground terminal located in power entrance box panel.

h. Measure continuity from pin 1 of power IN connector to pin 1 of power OUT connector; from pin 2 of power IN connector to pin 2 of power OUT connector; and from pin 4 of power IN connector to pin 4 of power OUT connector.

i. Measurements made by 4.6.1.1b, d, f, g and h shall indicate existence of electrical circuit continuity. (See 3.3.1.1)

4.6.1.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.3.1.2 shall be made on each power-line circuit indicated by the following paragraphs.

NOTE: Take preventive measures that will preclude personnel from coming into contact with conductors uder test during performance of ALL dielectric and insulation resistance test. Discharge each conductor to ground following each test, to eliminate danger of shock.

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a. Place all circuit controls listed in procedure 4.6.1.1a in the ON position.

b. Connect tester between pin 1 and pin 2 of power IN connector J1 in power entrance box. Adjust tester for a value of 500 volts DC and maintain this potential for a period of 30 seconds. The requirements of 3.3.1.2 shall be met.

c. Repeat procedure 4.6.1.2b with tester connected between pin 1 and pin 4 of power IN connector.

d. Repeat procedure 4.6.1.2b with tester connected between pin 2 and pin 4 of power IN connector.

4.6.1.3 Insulation resistance.- Insulation resistance tests which satisfy the requirements of 3.3.1.3, shall be made on each power-line circuit indicated by the following paragraph (Method 302 of MIL-STD-202 shall be employed using a megohmeter or equivalent tester):

a. Connect tester beween pin 1 and pin 2 of power IN connector in power entrance box. Adjust tester for a value of not less than 100 nor more than 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.1.3 shall be met.

b. Repeat procedure 4.6.1.3a with tester connected between pin 1 and pin 4 of power IN connector.

c. Repeat procedure 4.6.1.3a with tester connected between pin 2 and pin 4 of power IN connector.

NOTE: To eliminate danger of shock. Discharge conductors following each test by connecting a 10 to 60 ohm wirewound resistor (or equivalent test lamp) to the pins tested..

4.6.2 Power distribution and operational functions tests.- Power distribution and operational functions tests which satisfy the requirements of 3.3.2 shall be made as indicated by the following paragraphs:

4.6.2.1 Test preparation.- (See power schematic plate)

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a. Replace or reconnect all electrical loads and line voltmeter removed or disconnected by procedure 4.6.1a.

b. Replace power distribution box cover.

c. Place circuit breakers and switches in the OFF position.

d. Place a 115 volt AC test lamp with grounding type plug into each convenience and equipment receptacle.

e. Open (and secure open) outer exhaust port covers of blower.

f. Place by-pass blackout switch in the ON position.

g. Insert AC line plugs of exhaust blower into receptacle.

h. Insert AC line plug of heater into receptacle. Place heater switch in FAN position.

i. Insert AC line plug of air conditioner. Set the following controls:

(1) SW1 fresh air control to 0.

(2) SW2 ON-OFF switch to ON.

(3) SW3 HEAT-VENT-COOL switch to VENT.

(4) SW4 WARM-COOL switch to WARM.

4.6.2.2 *Line voltmeter and indicator lamp.-*

a. Apply 115 volt AC (nominal), 60 cycles, power to power IN connector in power entrance box. Power indicator neon lamp shall light; facility voltmeter in power distribution panel shall indicate power to the shelter.

b. Measure specific line voltage by connecting calibrated external AC voltmeter between pin 1 and pin 2 of power OUT connector.

c. Read voltage indicated on facility voltmeter. The voltage read shall be within 3% of the voltage read by 4.6.2.2b.

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d. Place main circuit breaker in ON position.

4.6.2.3 Circuit breakers and switches.-

Circuit No. 1.- (Lights circuit breaker)

a. Place circuit breaker in the ON position. Neon lamp shall light.

b. Place fluorescent light switch in ON position. All fluorescent lamps shall light.

c. Place incandescent light switch in ON position. Incandescent lamps shall light.

d. Place night light switch in ON position. Incandescent lamps shall light.

e. Place circuit breaker in OFF position. Neon lamp and all fluorescent and incandescent lamps shall go out.

f. Place circuit breaker in ON position. With shelter door open, place by-pass blackout switch in OFF position. All fluorescent and incandescent lamps shall go out.

g. Close shelter door. All fluorescent and incandescent lamps shall light.

h. Open door and place by-pass blackout switch in ON position.

Circuit No. 2.- (Heater circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light and heater fan shall operate and blow air.

b. Place thermostat control on heater to a high heating position. Place heater 1 in HEAT position. Ammeter shall indicate approximately 15 amps. After a few minutes the heater shall blow warm air. Check that the thermostat controls heater operation by turning thermostat to a colder temperature setting. When the lower temperature is reached, a click will be heard and the shelter's ammeter shall show a drop of about 13 amps. The fan shall continue to blow.

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c. Place circuit breaker in the OFF position. Neon lamp shall go out and heater shall cease to operate.

Circuit No. 3.- (Convenience and intercom circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light and test lamp in each receptacle shall light.

b. Place circuit breaker in OFF position. Neon lamp shall go out. Test lamp in each receptacle shall go out.

Circuit No.4.- (Blower circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light.

b. Place blower switch in ON position. Blower shall operate and exhaust port cover shall open.

c. Place circuit breaker in OFF position. Neon lamp shall go out and blower shall stop and exhaust port cover shall close.

Circuit No. 5.- (Air conditioner circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light and air conditioner blower shall operate and blow air.

b. Place the SW3 HEAT-VENT-COOL switch to HEAT. After a few minutes of operation, air conditioner shall blow warm air.

c. Place the SW4 WARM-COOL switch to COOL. Place SW3 HEAT-VENT-COOL switch to COOL. After a few minutes of operation, the air conditioner shall supply cool air.

d. Place the SW2 ON-OFF switch to OFF.

e. Place circuit breaker in OFF position neon lamp shall go out.

Circuit No. 6.- (Spare circuit breaker)

a. Place circuit breaker in ON position. Neon lamp shall light.

b. Place circuit breaker in OFF position. Neon lamp shall go out.

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Circuit No. 7.- (Main circuit breaker)

a. Place all circuit breakers in ON position. All lamps shall light. Blower, heater and air conditioner shall operate. Test lamp in each receptacle shall light.

b. Place circuit breaker in OFF position. All lamps shall go out. Blower heater and air conditioner shall cease to operate. Test lamp in each receptacle shall go out.

c. Disconnect all test equipment and secure operational test.

4.6.3 Signal (audio) system wiring and components installation tests.- (See 3.3.3)

4.6.3.1. Continuity of conductors.- Continuity resistance tests to meet the requirements of 3.3.3.1 shall be made on each audio circuit indicated by the following paragraphs (using a low resistance continuity tester or equivalent, producing a test current of not more than .010 amps):

a. From U-187( )/G of J1, J2 and J3

Pin 25B to end of wire blk (W4)
Pin 25A to end of wire red (W4)
Pin 26B to end of wire blk (W4A)
Pin 26A to end of wire red (W4A)

b. From binding posts signal entrance boxes, signal 1, signal 2 and signal 3:

E25B to E51B
E25B to E77B
E25A to E51A
E25A to E77A
E26B to E52B
E26B to E78B
E26A to E52A
E26A to E78A

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4.6.3.2 Dielectric strength of insulation.- Dielectric strength of insulation tests which satisfy the requirements of 3.3.3.2 shall be made for each audio circuit indicated by the following paragraphs (Method 301 of MIL-STD-202 shall be used utilizing a non-destructive dielectric test instrument and suitable test jigs or an automatic tester may be used subject to government approval):

a. Connect the following to facility ground terminal:

(1) Pin 1A thru 26B of audio connector J1.

(2) Pin 1A thru 26B of audio connector J2.

(3) Pin 1A thru 26B of audio connector J3.

b. Connect tester between facility ground terminal and pin 1A of audio connector J1. Remove ground connection form pin 1A of audio connector J1.

c. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds the requirements of 3.3.3.2 shall be met.

d. Disconnect tester lead from pin 1A of audio connector J1 and re-established ground connection of pin 1A of audio connector J1.

e. Repeat procedures 4.6.3.2b thru d for each remaining (pins 2B thru 26B of audio connector J1) each time ungrounding the pin of audio connector J1 correlating with the circuit being tested and reconnecting pin to facility ground terminal following completion of test.

f. Repeat procedures 4.6.3.2c thru e for each conductor terminated in audio connector J2 and J3.

NOTE: To eliminate danger of shock. Discharge conductors following each test by connecting a 500 ohm wirewound resistor (or equivalent test lamp) to the pins tested.

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4.6.3.3 Insulation resistance.- Insulation resistance which satisfy the requirements of 3.3.3.3 shall be made on each audio circuit indicated by the following paragraphs (Method 302 of MIL-STD-202 shall be used, utilizing a megohmmeter or equivalent tester, and suitable test jig or an automatic tester subject to government approval):

a. Repeat procedures 4.6.3.2a thru g excent 4.6.3.2c which shall be replaced by 4.6.3.3b below.

b. Adjust tester to 500 volts DC output. The requirements of 3.4.3.3 shall be met.

NOTE: To eliminate danger of shock. Discharge conductors following each test by connecting a 500 ohm wirewound resistor (or equivalent test lamp) to the pins tested.

4.7 System operation test.- (See 3.4)

4.7.1. Signal circuit preparation prior to conducting test.-

a. Install ten batteries BA-200/U in the battery compartments in power pack (PP-990( )/G).

b. Install two batteries BA-30 in the battery compartment located in the rear of the operators pack (TA-220( )/P).

c. Install two batteries BA-30 in the battery compartment located in the rear of the jack field section (TA-207( )/P).

d. Connect pair from the -24V (Blk wire) and +24V (Red wire) binding posts on the jack field section (TA-207( )/P)1 and 2 to the corresponding binding posts on operators pack (TA-220( )/P).

e. Connect cable W5 as follows:

(1) Two wires wht-blk from the RWG SUP binding posts on the power pack (PP-990( )/G) to the EXT GEN binding post on the operators pack (TA-220( )/P).

(2) One wire GREEN from the ST. VIB binding post on the power pack (PP-990( )/G) to the ST. VIB binding post on the operators PACK (TA-220( )/P).

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(3) Two wires blk wire from the -24V binding post on the power pack (PP-990( )/P) to the corresponding binding post on the operators pack (TA-220( )/P); red wire from the +24V binding post on the power pack (PP-990( )/P) to the corresponding binding post on the operators pack (TA-220( )/P).

f. Connect signal circuits to the line binding posts located in the rear of the jack field sections (TA-207( )/P) upper #1, lower #2 as shown on Table VI.

g. Connect green wire as follows:

(1) From earth ground binding post on the operators pack (TA-220( )/P) to EARTH GRD binding post on jack field section (TA-207( )/P) #2 lower.

(2) From earth ground binding post on jack field section (TA-207( )/P) #2 lower to EARTH GRD binding post on jack field section (TA-207( )/P) #1 upper.

(3) From earth ground binding post on the operators pack (TA-220( )/P) to ground terminal in power entrance panel.

h. Turn all line selector switches (located behind the designation strips) on Switchboard Signal Assemblies of TA-207( )/P 1&2 to M (Magneto).

i. Position MA switch to VIS on each TA-207( )/P.

j. Position lamps switch on each TA-207( )/P to OFF.

k. Position Talk Bat switch on switchboards to ON.

l. Position CIV TRKS switches on each TA-207( )/P to the OFF position.

m. Open top cover of Power Supply PP-990( )/G.

n. Position BATT switch to INT.

o. Position INT SWBD BATT switch to HI.

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p. Position and hold BATT check switch to 1. Meter shall indicate 20 to 26.5 volts. If meter indication is greater than 26.5 volts, position INT SWBD BATT switch to LOW. If meter indication is less than 20 volts position INT SWBD BATT switch to HI.

q. Position and hold BATT check switch to 2. Meter shall indicate 20 to 30 volts.

r. Release batt, check switch and close top cover of power supply.

s. Connect an operators Handset H-91( )/U to the receptacle on switchboard.

t. Place all cord circuit switches of SB-86( )/P Switchboard in their neutral (handle straight up) position.

u. Prepare two TA-312( )/PT Telephones for use during the tests by performing the following steps:

(1) Install two batteries BA-30 in each battery compartment.

(2) Turn CB-LB CRS screwdriver switch on all TA-312( )/PT Telephones to LB position.

(3) Turn EXT-INT switch to INT.

(4) Turn loudness control on all TA-312( )/PT telephones to LOUD.

4.7.2 Signal Circuit Test.- Signal 1 and Signal 2 Test.-

a. Connect one TA-312( )/PT Telephone to pair 1 binding post (E1) in signal entrance box. Call this Telephone 1.

b. Connect the second TA-312( )/PT Telephone to pair 1 binding post (E27) in signal entrance box. Call this Telephone 2.

NOTE: For purpose of this procedure the cord circuits of both switchboards shall be considered as being numbered consecutively from left to right.

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c. Lift the handset of TA-312( )/PT, Telephone 1, actuate ringer. Line signal #1 on TA-207( )/P upper #1 of SB-86( )/P shall turn from black to white and panel lamps shall light.

d. On operators keyshelf, position the cord circuit switch of cord circuit to talk-list position.

e. Place the answer cord plug (in line with the cord circuit switch just moved - top cord) into line jack #1 (directly below the line signal indicating white). Line signal turns from white to black and panel lamps extinguish.

f. Communication between switchboard operator and the operator of TA-312( )/PT, Telephone 1, shall be loud and clear.

g. Place the call cord plug of this cord circuit into line jack #1 on TA-207( )/P lower #2 of SB-86( )/P.

h. Place the ringing circuit switch to RING FWD. TA-312( )/PT, Telephone 2 shall ring.

i. Lift handset of TA-312( )/PT Telephone 2. Communication between TA-312( )/PT, Telephone 2, switchboard operator, and TA-312( )/PT Telephone 1, shall be loud and clear.

j. Place cord circuit switch of cord circuit #1 in neutral (handle straight up) position.

k. Actuate ringers of both TA-312( )/PT Telephones. Supervisory signals of the cord circuit being used shall turn from black to white.

l. Pull the plugs of both cords from the line jacks and return them to their plug seats. Supervisory signals shall turn from white to black.

m. Repeat steps a thru l using signal lines and cord circuits listed below:

NOTE: At any convenient point during the test, position the NA switches on Switchboard SB-86( )/P, to AUD. Actuation of ringers on the TA-312( )/PT Telephones shall cause the switchboard alarms to ring.

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SB-86( )/P Upper TA-207( )/P #1 Binding Post		SB-86( )/P Lower TA-207( )/P #2 Binding Post		Cord Circuit
Line 2	(E2)	Line 2	(E28)	2
Line 3	(E3)	Line 3	(E29)	3
Line 4	(E4)	Line 4	(E30)	4
Line 5	(E5)	Line 5	(E31)	5
Line 6	(E6)	Line 6	(E32)	6
Line 7	(E7)	Line 7	(E33)	7
Line 8	(E8)	Line 8	(E34)	8
Line 9	(E9)	Line 9	(E35)	9
Line 10	(E10)	Line 10	(E36)	10
Line 11	(E11)	Line 11	(E37)	11
Line 12	(E12)	Line 12	(E38)	12
Line 13	(E13)	Line 13	(E39)	13
Line 14	(E14)	Line 14	(E40)	14
Line 15	(E15)	Line 15	(E41)	15
Line 16	(E16)	Line 16	(E42)	16
Line 17	(E17)	Line 17	(E43)	any
Line 18	(E18)	Line 18	(E44)	any
Line 19	(E19)	Line 19	(E45)	any
Line 20	(E20)	Line 20	(E46)	any
Line 21	(E21)	Line 21	(E47)	any
Line 22	(E22)	Line 22	(E48)	any
Line 23	(E23)	Line 23	(E49)	any
Line 24	(E24)	Line 24	(E50)	any

4.7.3 Signal Circuit Test Signal 3.-

a. Transfer TA-312( )/PT Telephone 1 from signal 1 binding post to signal 3 binding post.

b. Transfer TA-312( )/PT Telephone 2 from signal 2 binding post to signal 3 binding post.

c. Repeat procedures 4.7.2 thru 1 using the signal lines and cord circuits listed below:

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SB-86( )/P Upper TA-207( )/P Binding Post	SB-86( )/P Lower TA-207( )/P Binding Post
Line 25 (E53)	Line 25 (E59)
Line 26 (E54)	Line 26 (E60)
Line 27 (E55)	Line 27 (E61)
Line 28 (E56)	Line 28 (E62)
Line 29 (E57)	Line 29 (E63)
Line 30 (E58)	Line 30 (E64)

4.7.4 Phone, Intercom Tests.-

a. Connect TA-312( )/PT Telephone 1 to binding post E25, in signal 1 entrance box.

b. Connect TA-312( )/PT Telephone 2 to end of cable W4 in shelter.

c. Lift the handset of TA-312( )/PT Telephone 1, actuate ringer. The TA-312( )/PT Telephone 2 shall ring.

d. Lift the handset of TA-312( )/PT Telephone 2, communication between Telephone 1 and Telephone 2 shall be loud and clear.

e. Connect an LS-147( )/FI to #1 binding post E26 of signal 1 entrance box.

f. Connect an LS-147( )/FI to #2 end of cable W4A in shelter.

g. Establish voice communications between LS-147( )/FI #1 and #2.

4.8 Airtightness test.- (See 3.5)

a. The air tightness shall be tested by introducing clean dry air into the closed shelter through an existing appropriate opening (i.e. drain fixture) to create an internal air pressure that exceeds the external by a minimum of 6 inches of water.

b. A manometer graduated 0.1 inches or less shall be utilized to measure this differential pressure.

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c. The test apparatus shall include an air supply sufficient to maintain the required air pressure, an airflow regulator and an input air pressure gage.

d. The shelter facility shall be tested for air tightness by painting the sealed joints, seams, rivets, etc. of the external modifications with Leak-Tac Formula #372 (American Gas & Chemicals, Inc., Leak-Tac Division) or equal soap and water solution.

e. To minimize air loss, temporary seals, shims and tape are permitted to seal the shelter door, blower covers and other through opening not being tested.

f. For the test to function properly, openings are required through the inside wall to the center of the hat sections in each rear, front and curbside panel where external modifications have been made on the shelter. These openings to the hat sections are obtained by drilling one-sixteenth holes. Ten holes will be drilled in locations specified by the government technical representative.

g. There shall be no leakage of the areas being tested as evidenced by bubbling or spurting of the test solution.

4.9 Rail transport, impact test.- The shelter facility loaded to simulate the AN/TTC-29 using live and dummy loads, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown slings, shall be as specified in Figure 1 and shall meet the provisions of the test as specified in paragraph 4.9.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in paragraph 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptalbe to the government technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. The test shall consist of four impacts. Two impacts with the shelter facility positioned longitudinally on the flatcar, one from each end of the shelter facility and two impacts with the shelter facility positioned transversely on the flatcar,

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one from each side of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After dach impact the shelter facility shall meet the requirements of paragraph 3.6.

4.9.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall not be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.9.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

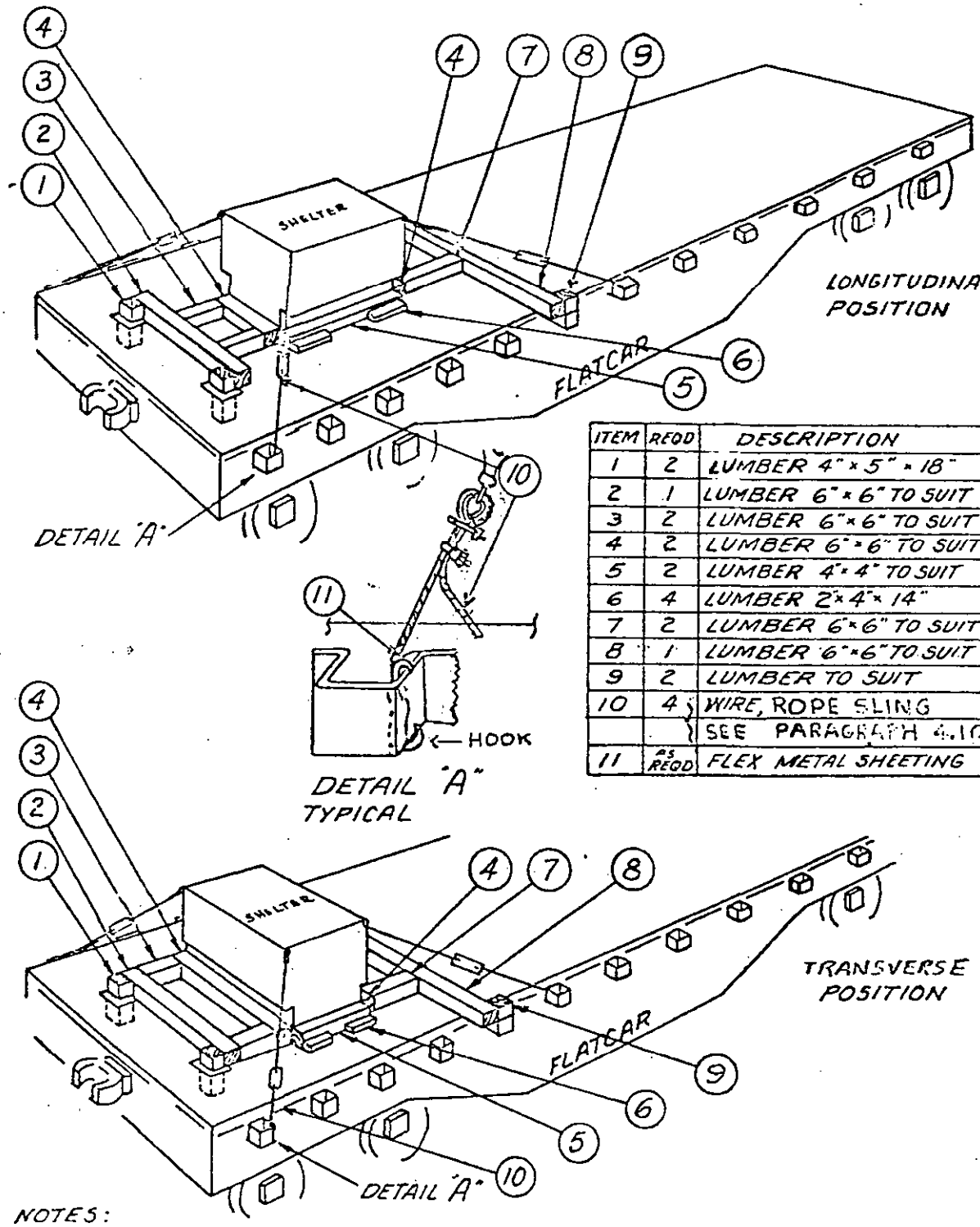
a. A coarse size of either coal, crushed stone, gravel, crushed concrete or metal scrap (other than sheet metal, metal shavings or other springly forms of metal).

b. Chunk, slap, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.10 Drop test.-

4.10.1 Flat drop.- The shelter facility loaded as in 4.9 shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

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ITEM	REQD	DESCRIPTION
1	2	LUMBER 4" × 5" × 18"
2	1	LUMBER 6" × 6" TO SUIT
3	2	LUMBER 6" × 6" TO SUIT
4	2	LUMBER 6" × 6" TO SUIT
5	2	LUMBER 4" × 4" TO SUIT
6	4	LUMBER 2" × 4" × 14"
7	2	LUMBER 6" × 6" TO SUIT
8	1	LUMBER 6" × 6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE, ROPE SLING SEE PARAGRAPH 4.10
11	AS REQD	FLEX METAL SHEETING

NOTES:

1- THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS.

2- THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR.

FIGURE #1

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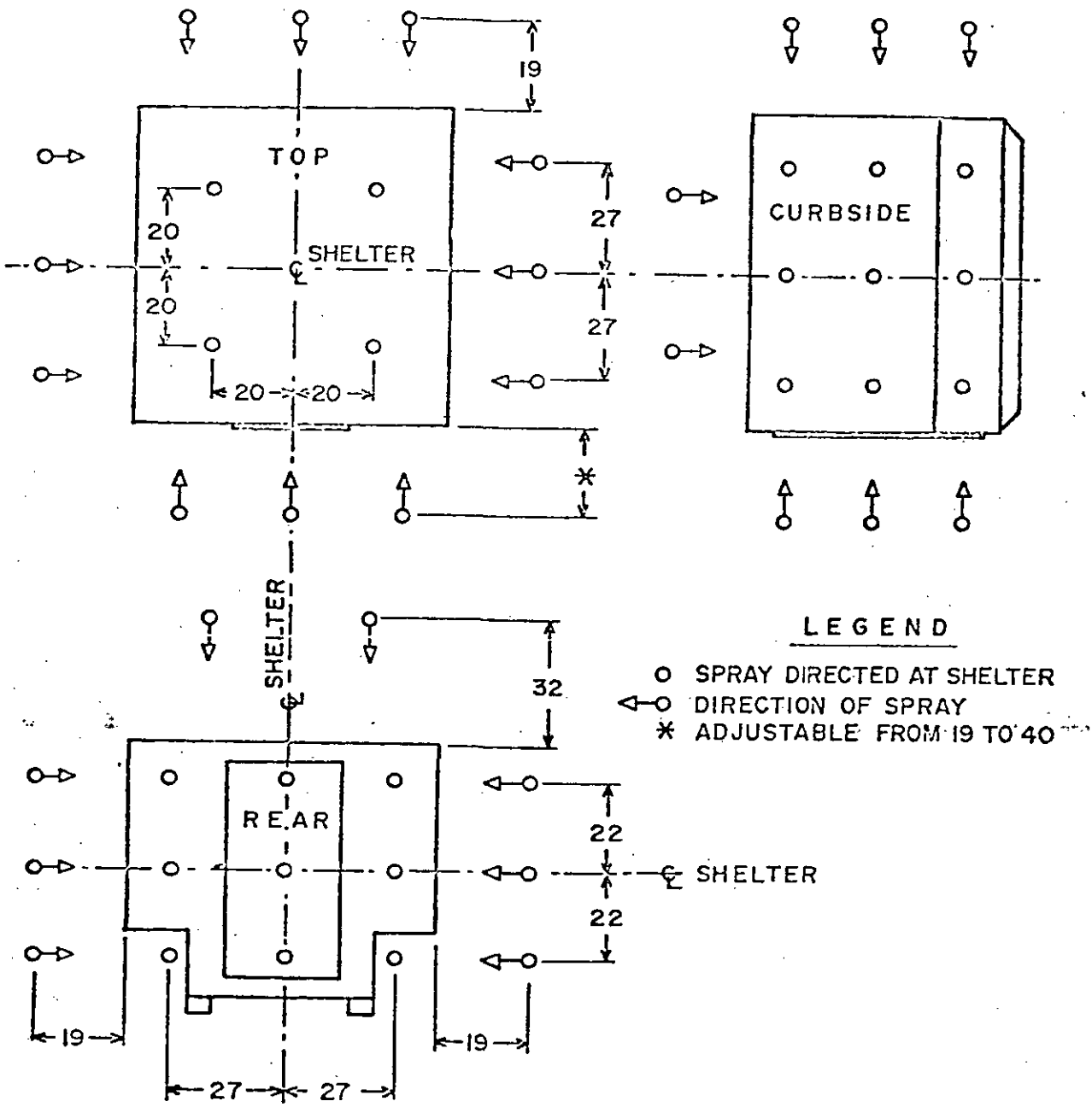
4.10.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.11 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test (NOTE: Perform tests with inside covers of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzels, model G 29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 2 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.12, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position (with exception of the exhaust blower vent covers at the front of the shelter, the door vent panel at the rear of the shelter and air conditioning duct port covers on the curbside wall) and repeat a above.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER SURFACE.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE #2

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4.12 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 26 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.13 Mounting insert test.-

4.13.1 First article and Group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejecttion of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.13.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V - Mounting Inserts Loads

Insert size	First Article and Group C Test		Production Test	
	Axial(lbs)	Torque(in.-lbs)	Axial(lbs)	Torque(in.-lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.14 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

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a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks, patch panels and cabinets.

4.16 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

TABLE VI - Classification of Visual & Mechanical Defects

Classification		Defects
Major	1.	Mounting plates missing or mislocated.
	2.	Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.
	3.	Welds - cracked or porous.
	4.	Sealer improperly applied (holes, seperations, or lack of adhesion).
	5.	Mounting inserts missing or mislocated.
	6.	Burrs or sharp edges in wire duct not removed.
	7.	Equipments or bracketry loose or missing.
	8.	Doors, etc. inoperative.
	9.	Electrical receptacles, switches, connectors, wire ducts or other electrical components loose, improperly located or inoperative.
	10.	Locking or holding devices missing or inoperative.
	11.	Gaps between section or wire duct exceed 1/16".
	12.	Gaskets, missing.
Minor	1.	Doors, not easily operable.
	2.	Fastening devices difficult to operate.
	3.	Finish - abrasions or scratches.
	4.	Burrs and sharp edges on bracket not removed.
	5.	Screws loose.
	6.	Gaskets, completely and securely adhered.

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4.18 Quality conformance inspection for packaging.- Quality conformance inspection for packaging shall be in accordance with Group A, Group B and, when required (See 6.2g), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.19 Rough handling test (packaging).- When rough handling test in accordance with MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.6	Electrical installation tests
Paragraph 4.16	Visual and mechanical inspection

5. PACKAGING

5.1 Packaging requirements.- Packaging shall be in accordance with applicable requirements of MIL-S-55507.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-392( )/TTC-29 is the shelter facility portion of the AN/TTC-29 and it is used to house the operational equipment including:

a. One each SB-86( )/P Switchboard, Telephone Manual.

b. One each TA-207( )/P Signal Assembly Switchboard.

c. One each TA-312( )/PT Telephone.

d. One each LS-147( )/FI Intercom.

e. One each 1500 watt Heater.

f. One each 6000 BTU Air Conditioner.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title number and date of this specification and any amendment thereto.

b. First article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads for the equipments (1 each) and government furnished live loads (Intercom

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and telephone component) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-392( )/TTC-29( ) on order.

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

f. Level A or level B preservation and packing (See Section 5).

g. When the rough handling test is required. (See Section 4).

6.3 Inspection.- Inspection is the examination, testing or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.4)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-392R/TTC-29. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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